An Efficient Framework for Phishing URL Detection Using Adaptive Drift-Aware Feature Monitoring and Online Learning

Implementation and evaluation — CS 552 Group Project

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Assigned paper — full reference

Sruthi Krishnan and Manohar Naik S. "An Efficient Framework for Phishing URL Detection Using Adaptive Drift-Aware Feature Monitoring and Online Learning." *Security and Privacy*, vol. 8, no. 4, article e70057, 2025. John Wiley & Sons Ltd. Department of Computer Science, Central University of Kerala, Periye, India.
DOI: 10.1002/spy2.70057
`https://onlinelibrary.wiley.com/doi/full/10.1002/spy2.70057`

Task and paper idea

Task: implement the methodology of the paper and reproduce its experiments as closely as possible.

- Phishing URLs change over time (concept drift); a model trained once goes stale.
- Paper: detect from the **URL string only** (no HTML, no third-party lookups).
- Offline: LASSO + mutual information pick the feature set F^*; SGD classifier trained.
- Online: per-feature KS test watches mini-batches; on drift, F^* is re-selected and the model is updated incrementally, with no full retraining.

Pipeline as implemented

Paper component	**Implementation**
Lexical features (Table 3)	58 URL-string features, `features/extractor.py`
LASSO + MI, Alg. 1	$\lambda = 10^{-4}$, $S = w + l$, $\tau = 75$th percentile, history H
KS drift monitor	$\delta = 0.05$, buffer 1500, $B = 500$, per-feature test
Online learner (Table 4)	SGD-SVM / LR / Perceptron, $\eta_0 = 0.01$, adaptive LR, balanced, early stopping
Framework loop	test-then-train; drift $\Rightarrow$ re-select F^* $\Rightarrow$ partial retraining

16 unit tests; every experiment is one `make` target.

Data and protocol

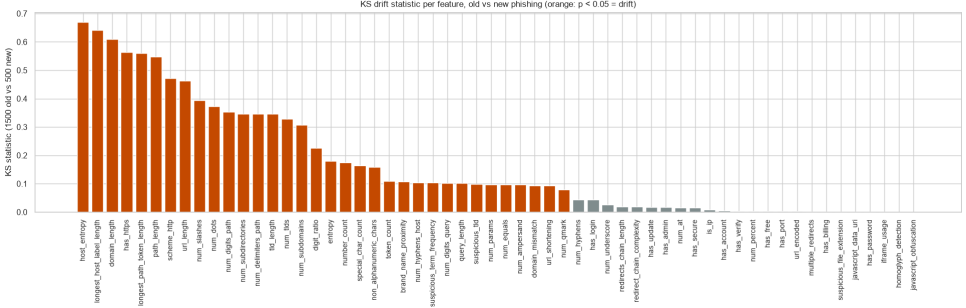
- **PhiUSIIL** (235k), **Kaggle** "Phishing Site URLs" (21k balanced), **ISCX-URL-2016** (authors' 79 lexical features, 15.4k), **Hannousse 2021** (11.4k URLs).
- **Combined corpus:** PhiUSIIL + Kaggle + Hannousse, de-duplicated (263k).
- **New URLs:** live PhishTank feed restricted to the paper's window (Jun 2024 – Jan 2025): 12,888 phishing URLs, streamed chronologically with held-out benign URLs.
- 80:20 split, seed 42; online phase uses prequential (test-then-train) evaluation, 5 seeds.
- Positive class = phishing.

Offline phase (paper Tables 5–7, Fig. 6)

Corpus		Acc.	Prec.	Rec.	F1
PhiUSIIL	this study	99.38	99.99	98.54	99.26
	paper	98.78	97.84	98.73	98.28
Kaggle	this study	73.81	81.84	60.33	69.46
	paper	99.00	98.92	98.77	98.85
ISCX	this study	90.83	90.49	90.97	90.73
	paper	99.02	98.39	99.63	99.01
Combined	this study	93.58	91.78	93.33	92.55
	paper	99.85	—	—	—

✓ SGD-SVM is the best SGD variant (as in the paper); PhiUSIIL matches.
All ISCX features: 96.1%; random forest on the same F^*: 98.3% — level depends on features and corpus version.

Feature drift (paper Figs. 4, 5, 7)



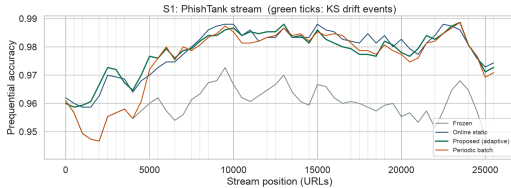
34 of 58 features drift between old and new phishing URLs ($\delta = 0.05$); largest: host entropy, host label length, domain length, scheme, path length, URL length.

Online phase under drift

PhishTank stream	Acc.	F1
Frozen offline	92.63	92.03
Online, static F^*	97.77	97.79
Proposed	**97.80**	**97.82**
Periodic batch	97.31	97.33

✓ Online learning: +5.1 points over frozen, low variance.

✓ Cross-corpus shift: frozen 55.5%, static 70.5%, proposed 73.2%.



Adaptive vs. static feature selection (paper Table 9)

F1 (%)	100 URLs		500 URLs		1000 URLs	
	static	adaptive	static	adaptive	static	adaptive
This study	96.19	96.08	96.25	96.90	96.54	96.85
Paper	92.31	96.22	93.89	97.15	93.73	98.30

✓ Adaptive F1 within 0.1–1.5 points of the paper.

✓ At $N \geq 500$: accuracy $+0.7$ point, FPR 6.8% $\rightarrow$ 5.4%; up to $+12$ points accuracy under a larger shift.

Our static baseline is stronger than the paper's, so the static–adaptive gap is smaller.

Runtime (paper Tables 10–11)

	Incremental	Batch retrain	Gain
Model update, 100–1000 URLs	1.5–1.9 ms	$\approx$4 s	$>$2000$\times$
Feature adaptation	213 ms	2.7 s	12.9$\times$

Per-URL latency (extract + score, 1 CPU thread): **0.16 ms** (paper: 12.4–15.2 ms; different hardware).

✓ Same conclusion as the paper: incremental updating is far cheaper than retraining.

Table of evaluations

Requirement / paper claim	Status	Evidence
Implement the proposed methodology	✓	all components, end-to-end
Source code, implementation/results, report, presentation/demo	✓	repo, `RESULTS.md`, PDF, `web/` demo
Full title and complete reference on slide 1	✓	slide 1
SGD-SVM best SGD variant; PhiUSIIL accuracy (Tab. 7)	✓	99.38% vs. 98.78%
Feature distributions drift (Figs. 4, 7); feature set evolves (Tab. 8)	✓	34/58 features; per-event changes
Online learning beats frozen model	✓	$+5.1$ / $+15.0$ points
Adaptive features, F1 at 100/500/1000 URLs (Tab. 9)	✓	96.08 / 96.90 / 96.85 vs. 96.22 / 97.15 / 98.30
Incremental vs. batch cost; real-time latency (Tabs. 10–11)	✓	$>2000\times$; 0.16 ms
Kaggle / ISCX / combined $\approx$99% (Tabs. 5, 6, Fig. 6)	~	73.8 / 90.8 / 93.6%; same model ranking, lower level

Conclusions

- The full method is implemented and every experiment of the paper's results section has been run.
- Qualitative findings hold: SGD-SVM is the best linear learner, features drift, drift-triggered re-selection with online learning keeps accuracy where a frozen model degrades, and incremental updates are orders of magnitude cheaper than retraining.
- Absolute accuracy on Kaggle, ISCX and the combined corpus depends on the corpus versions and feature set (e.g. 90.8% $\rightarrow$ 96.1% on ISCX with all features); PhiUSIIL matches the paper.
- Limits: URL-only features; drift aggregation rule, history weight and replay epochs are our choices where the paper is silent.